

Worked Examples

Complex Arithmetic

Ex 1: Simplify $(1-i)^{12}$

Step 1: Convert to polar

- $|1-i| = \sqrt{1^2+1^2} = \sqrt{2}$
- Angle: $\theta = -\pi/4$ (4th quadrant)
- So $1-i = \sqrt{2}e^{-i\pi/4}$

Step 2: Apply power rule

- $(1-i)^{12} = (\sqrt{2})^{12} \cdot e^{-12i\pi/4}$
- $= 2^6 \cdot e^{-3i\pi} = 64 \cdot e^{-3i\pi}$

Step 3: Convert back

- $e^{-3i\pi} = e^{-i\pi} = -1$
- Final: $64 \cdot (-1) =$ **-64**

Ex 2: Complex product $(1/2 + \sqrt{3}/2 \cdot i)(1/2 - \sqrt{3}/2 \cdot i)$

Method 1 - Conjugate Pattern:

- Pattern: $(a+bi)(a-bi) = a^2 + b^2$
- Here: $a = 1/2, b = \sqrt{3}/2$
- Result: $(1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 =$ **1**

Method 2 - FOIL (verify):

- Real part: $(1/2)(1/2) - (\sqrt{3}/2)(-\sqrt{3}/2)$
- $= 1/4 + 3/4 = 1$
- Imaginary: $(1/2)(-\sqrt{3}/2) + (\sqrt{3}/2)(1/2) = 0$
- Result: $1 + 0i =$ **1**

Ex 3: Rotate $-1/2 + \sqrt{3}/2 \cdot i$ **by** $-\pi/6$

Step 1: Identify original angle

- This is $e^{i2\pi/3}$ (120 degrees)
- Verify: $\cos(2\pi/3) = -1/2, \sin(2\pi/3) = \sqrt{3}/2$

Step 2: Apply rotation

- Multiply by $e^{-i\pi/6}$ to rotate clockwise
- $e^{i2\pi/3} \cdot e^{-i\pi/6} = e^{i(2\pi/3-\pi/6)}$
- $= e^{i(4\pi/6-\pi/6)} = e^{i\pi/2}$

Step 3: Convert to rectangular

- $e^{i\pi/2} = \cos(\pi/2) + i\sin(\pi/2) = 0 + i =$ **i**

Polynomial Factoring

Factor $x^3 + 1$ **and find all roots**

Step 1: Use sum of cubes formula

- $x^3 + 1 = (x+1)(x^2 - x + 1)$
- First root: $x = -1$

Step 2: Quadratic formula on $x^2 - x + 1 = 0$

- $x = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2}$
- $= \frac{1 \pm i\sqrt{3}}{2}$

Step 3: All three roots

- Root 1: $x = -1$
- Root 2: $x = \frac{1+i\sqrt{3}}{2} = e^{i\pi/3}$
- Root 3: $x = \frac{1-i\sqrt{3}}{2} = e^{-i\pi/3}$
- These are the three cube roots of -1

Aliasing

Find alias: $f_s = 44100$ **Hz, signal = 32100 Hz**

Step 1: Check Nyquist

- Nyquist freq $= f_s/2 = 22050$ Hz
- $32100 > 22050$, so aliasing occurs

Step 2: Calculate alias frequency

- Use: $f_{alias} = |f - nf_s|$ where n minimizes
- $n = 0$: $|32100 - 0| = 32100$ Hz
- $n = 1$: $|32100 - 44100| = 12000$ Hz (smaller!)
- $n = 2$: $|32100 - 88200| = 56100$ Hz (larger)

Step 3: Verify and conclude

- 12000 Hz is in valid range $[0, 22050]$
- The 32100 Hz signal appears as **12000 Hz**

DFT Calculation

Find $X[1]$ **for** $x = [1, 0, 1], N = 3$

Step 1: Write DFT formula

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}$$

Step 2: Substitute values ($k = 1, N = 3$)

- $X[1] = x[0]e^0 + x[1]e^{-i2\pi/3} + x[2]e^{-i4\pi/3}$
- $= 1(1) + 0(e^{-i2\pi/3}) + 1(e^{-i4\pi/3})$
- $= 1 + e^{-i4\pi/3}$

Step 3: Simplify the exponential

- $e^{-i4\pi/3} = \cos(-4\pi/3) + i\sin(-4\pi/3)$
- $= \cos(4\pi/3) - i\sin(4\pi/3)$
- $= -1/2 - i \cdot (-\sqrt{3}/2) = -1/2 + i\sqrt{3}/2$
- Step 4:** Final answer
- $X[1] = 1 + (-1/2 + i\sqrt{3}/2) = 1/2 + i\sqrt{3}/2$
- This equals $e^{i\pi/3}$

- DFT coefficient: $c_1 = X[1]/N =$

FIR Filter Analysis

Analyze $y_t = x_t - (1/4)x_{t-1}$

Step 1: Transfer function

- Direct: $H(z) = 1 - (1/4)z^{-1}$

Step 2: Find zeros and poles

- Zero: Set $1 - (1/4)z^{-1} = 0$
- Solve: $z^{-1} = 4$, so $z = 1/4$
- Pole: At $z = 0$ (from z^{-1} term)
- Zero at $z = 1/4$, pole at $z = 0$**

Step 3: Frequency response at $\omega = 0$

- $H(0) = 1 - (1/4)e^{-i \cdot 0} = 1 - 1/4 =$ **3/4**
- DC gain is 0.75 (attenuation)

Step 4: Magnitude at $\omega = \pi/2$

- $H(\pi/2) = 1 - (1/4)e^{-i\pi/2}$
- $e^{-i\pi/2} = -i$, so $H(\pi/2) = 1 - (1/4)(-i) = 1 + i/4$

- $|H(\pi/2)| = |1 + i/4| = \sqrt{1 + 1/16} =$

IIR Filter with Impulse

Analyze $y_t = x_t - (1/4)y_{t-1}$, **impulse response**

Step 1: Find transfer function

- Rearrange: $y_t + (1/4)y_{t-1} = x_t$
- Z-transform: $Y(z)[1 + (1/4)z^{-1}] = X(z)$
- $H(z) = 1/[1 + (1/4)z^{-1}] = z/(z + 1/4)$

Step 2: Check stability

- Pole at $z = -1/4$
- $|pole| = |-1/4| = 1/4 < 1$ **STABLE**

Step 3: Impulse response calculation

- Input: $x = [1, 0, 0, \dots]$
- $y_0 = 1 - (1/4)(0) = 1$

- $y_1 = 0 - (1/4)(1) = -1/4$
- $y_2 = 0 - (1/4)(-1/4) = 1/16$
- $y_3 = 0 - (1/4)(1/16) = -1/64$

Step 4: Pattern recognition

- $h[n] = (-1/4)^n$ for $n \geq 0$
- Alternating sign, exponential decay
- Sum: $\sum h[n] = 1/(1 - (-1/4)) = 4/5$

Reson Filter Design

Design reson: $\omega = \pi/4$, **bandwidth** $B = 0.1$

Step 1: Calculate pole radius from bandwidth

- Narrow band formula: $B \approx 2(1 - R)$
- $0.1 = 2(1 - R)$, so $1 - R = 0.05$
- $R = 0.95$

Step 2: Place complex conjugate poles

- Poles at $z = 0.95e^{\pm i\pi/4}$
- In rectangular: $0.95 \cdot (\frac{1}{\sqrt{2}} \pm i \frac{1}{\sqrt{2}})$
- $= \frac{0.95}{\sqrt{2}}(1 \pm i) = 0.672(1 \pm i)$

Step 3: Write transfer function

- $H(z) = \frac{1}{1 - 2R \cos(\pi/4)z^{-1} + R^2 z^{-2}}$
- $\cos(\pi/4) = 1/\sqrt{2}$, so $2R \cos(\pi/4) = 2(0.95)/\sqrt{2} = 1.343$
- $H(z) = \frac{1}{1 - 1.343z^{-1} + 0.9025z^{-2}}$

Step 4: Calculate peak gain

- At resonance ($\omega = \pi/4$):
- Approximate gain $\approx 1/(1 - R) = 1/0.05 = 20$
- In dB: $20 \log_{10}(20) = 26 \text{ dB}$

Plucked String Tuning

Tune to 440 Hz, $f_s = 44100 \text{ Hz}$

Step 1: Calculate ideal period in samples

- $P = f_s/f_0 = 44100/440 = 100.227$ samples
- Need to achieve this total delay

Step 2: Allocate delays

- Try integer delay $L = 100$
- Low-pass filter adds 0.5 samples
- Total so far: $100 + 0.5 = 100.5$
- Still need: $100.227 - 100.5 = -0.273$ (negative!)

Step 3: Adjust allocation

- Use $L = 99$ instead
- With LP: $99 + 0.5 = 99.5$
- Need from all-pass: $100.227 - 99.5 = 0.727$

Step 4: Calculate all-pass coefficient

- For delay $\delta = 0.727$:
- $a = (1 - \delta)/(1 + \delta) = 0.273/1.727 = 0.158$

Step 5: Verify total delay

- Integer: 99 samples
- Low-pass: 0.5 samples
- All-pass: 0.727 samples
- Total: $99 + 0.5 + 0.727 = 100.227$ Perfect!

All-Pass Identification

Is $H(z) = (0.5z + 1)/(z + 0.5)$ **all-pass?**

Step 1: Convert to standard form

- Divide num and denom by z :
- $H(z) = (0.5 + z^{-1})/(1 + 0.5z^{-1})$

Step 2: Check all-pass structure

- All-pass form: $H(z) = (a + z^{-1})/(1 + az^{-1})$
- Our filter: numerator has $a = 0.5$
- Denominator also has $a = 0.5$
- Structure matches! YES, all-pass

Step 3: Verify magnitude = 1

- At any frequency: $|H(e^{i\omega})| = |(0.5 + e^{-i\omega})/(1 + 0.5e^{-i\omega})|$
- Numerator and denominator have same structure
- Can show $|H(e^{i\omega})| = 1$ for all ω

Step 4: Phase delay property

- All-pass changes phase without changing magnitude
- Used for fractional delay adjustment
- Phase: $\phi(\omega) = -2 \arctan(\frac{0.5 \sin \omega}{1 + 0.5 \cos \omega})$